

Introduction — Why Materials Matter

Technology is not only about machines or computers.
Everything around us is made of materials.

A hospital bed, a syringe, gloves, a chair —
all of these things are made of different materials.

- Because materials decide how things behave.
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- For example:
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- Metal is strong.
- Rubber is flexible.
- Plastic is light.
- Glass is hard but can break.

- If we choose the wrong material, the object will not work well.
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- A bed can break.
- Gloves can tear.
- A tube can crack.
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- In real hospitals, this is very important.
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- Doctors and patients depend on equipment.
- If materials are wrong, it can be dangerous.

- That is why engineers must understand materials.
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- They must ask:
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- What material is best?
- Why is it used here?
- What happens if we change it?

- Today, we will learn how to think about materials.
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- Not in a complex way —
- but step by step, using simple ideas.
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- Material → Property → Use
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- This is the key idea for today's lesson.

- **Game: Good Idea or Crazy Idea?**

- ^{٦٨} **Silk hospital bed sheets → Good Idea** Because silk is soft and comfortable for patients who stay in bed for a long time.
- ^{٦٨} **Stone hospital chair → Crazy Idea** Because stone is too heavy and very uncomfortable to sit on.
- ^{٦٨} **Foam rubber surgical knife → Crazy Idea** Because foam rubber is too soft and cannot cut anything.
- ^{٦٨} **Helium balloons as IV stands → Crazy Idea** Because helium balloons can fly away and are not stable.
- ^{٦٨} **Coconut shell as a medicine cup → Crazy Idea** Because coconut shell is difficult to clean and not safe for medicine.
- ^{٦٨} **Sponge mattress for very heavy patients → Crazy Idea** Because sponge is too soft and will break under heavy weight.
- ^{٦٨} **Cardboard wheelchair → Crazy Idea** Because cardboard is weak and will get wet and fall apart.
- ^{٦٨} **Jelly hospital floor → Crazy Idea** Because jelly is slippery and dangerous — people will fall.
- ^{٦٨} **Feather surgical gloves → Crazy Idea** Because feathers are too soft and cannot protect the doctor's hands.
- ^{٦٨} **Magnetic metal patient gown → Crazy Idea** Because magnetic metal can attract other metal objects and be dangerous.
- ^{٦٨} **Ice pack as a permanent hospital pillow → Crazy Idea** Because ice will melt and become wet and cold.
- ^{٦٨} **Paper surgical mask → Crazy Idea** Because paper gets wet easily and cannot filter air well.
- ^{٦٨} **Wooden stethoscope → Crazy Idea** Because wood cannot carry sound well and is difficult to clean.
- ^{٦٨} **Cotton candy oxygen tube → Crazy Idea** Because cotton candy will melt and is not safe for oxygen.
- ^{٦٨} **Gold hospital bed for VIP patients → Crazy Idea** Because gold is very expensive and too soft for a hospital bed.

- **Wooden syringe**
- **Paper hospital bed**
- **Glass gloves**
- **Chocolate IV tube**
- **Sponge wheelchair wheels**
- **Cotton knife**
- **Ice hospital chair**
- **Cardboard oxygen mask**
- **Rubber hammer**
- **Metal pillow**
- **Glass floor in the operating room**
- **Banana peel bandages**

- **Wooden syringe** → Bad Because wood is not clean and can break inside the patient.
- **Paper hospital bed** → Bad Because paper is weak and will tear when the patient lies down.
- **Glass gloves** → Bad Because glass is hard and will break easily, cutting the doctor's hands.
- **Chocolate IV tube** → Bad Because chocolate will melt and is not safe.
- **Sponge wheelchair wheels** → Bad Because sponge is too soft and the wheelchair won't move properly.
- **Cotton knife** → Bad Because cotton is soft and cannot cut anything.
- **Ice hospital chair** → Bad Because ice will melt and the patient will fall.
- **Cardboard oxygen mask** → Bad Because cardboard cannot hold oxygen and will get wet.
- **Rubber hammer** → Bad Because rubber is too soft and cannot hammer nails.
- **Metal pillow** → Bad Because metal is hard and uncomfortable for the patient to sleep on.
- **Glass floor in the operating room** → Bad Because it can break and is very dangerous.
- **Banana peel bandages** → Bad Because banana peel is slippery and not clean.

